

Neuro-Oracle: A Trajectory-Aware Agentic RAG Framework for Interpretable Epilepsy Surgical Prognosis

Longitudinal MRI trajectories · case retrieval · an LLM that explains its verdict

Aizierjiang Aiersilan* Mohamad Koubeissi

The George Washington University

*Corresponding author: Aizierjiang Aiersilan | alexandera@gwu.edu

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Motivation

Why surgical prognosis matters

Epilepsy affects about **50 million** people worldwide [19].

- Medication controls seizures in about **70%** of patients.
- The other **30%** are **pharmacoresistant**: surgery is the main curative route [18].
- Outcomes differ by surgery type [16, 7]:

Resection	Seizure-free at 2 yr
Temporal lobe	65–70%
Non-temporal / multilobar	40–55%

The clinical question

 Which patients are most likely to benefit from surgery?

Even initial responders can relapse later [4], so a reliable prognostic tool has real value.

Three obstacles for deep learning

1 Static scans

A single scan is **one moment in time**. It cannot show the **trajectory** of anatomical change.

2 Small cohorts

Surgical cohorts rarely exceed a **few hundred** patients, so large models **overfit** [14].

3 Black-box output

A bare probability gives **no verifiable rationale** for a surgical decision.

Our question

Can we predict outcome from the **change between scans**, and **explain** each prediction with **similar past cases**?

Where prior work stands

- **Static deep models:** 3D CNNs [9] and 3D ViTs [6] on one scan. ✗ Discard the longitudinal signal.
- **Longitudinal MRI:** Siamese networks [3] and deformation morphometry [1].
✗ Deformable warps can smooth away local change.
- **Retrieval-augmented generation (RAG)** [12]: grounds an LLM in retrieved evidence [20]. ✗ To our knowledge, not yet used with 3D longitudinal trajectories for epilepsy.

Our stance

Treat each patient as a **trajectory**, rigidly aligned and embedded, then let an LLM agent **reason over retrieved look-alike cases**.

1. ★ **Trajectory encoding** — a 3D Siamese contrastive encoder turns a pre/post-operative MRI pair into a **512**-dim vector Δv .
2. ★ **Agentic retrieval** — **Neuro-Oracle** retrieves the **5** most similar past trajectories and a quantized **Llama-3-8B** agent writes a grounded verdict.
3. ★ **Reproducible evaluation** — **7** methods on public **EPISURG** ($N=268$) with open-source tooling only.

Method

Framework overview

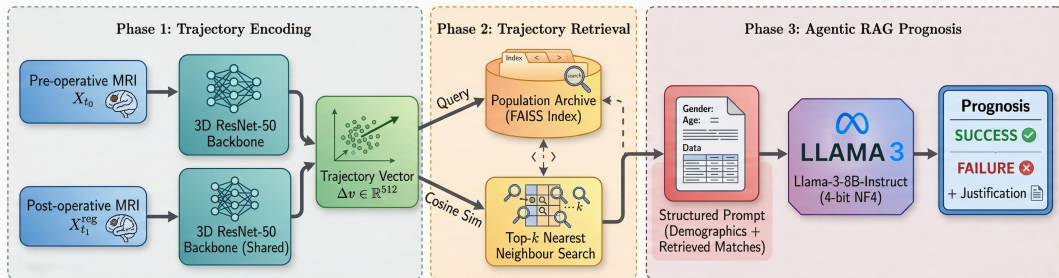


Figure 1: Neuro-Oracle in three phases: (i) encode the MRI pair into a trajectory vector; (ii) retrieve similar past cases; (iii) an LLM agent reasons over them to give a verdict and a justification.

→ Each phase can be **trained and deployed independently**.

Phase 1: contrastive trajectory encoding

- Post-op scan **rigidly registered** (6-DOF) to the pre-op scan — no deformable warp.
- One **shared 3D ResNet-50** backbone encodes both scans [2].
- The feature **difference** is projected and normalised.

Inputs: **128**³ voxels at **1** mm isotropic, z-score normalised. MLP: **2048** → **512** → **512**.

Trajectory vector

$$v_{t_0} = f_{\theta}(X_{t_0}), \quad v_{t_1} = f_{\theta}(X_{t_1}^{\text{reg}})$$

$$\Delta v = \ell_2\text{-norm}(\text{MLP}(v_{t_1} - v_{t_0})) \in \mathbb{R}^{512}$$

Phase 1: training objective

The encoder learns to pull same-class trajectories together while handling a 4:1 class imbalance.

Combined loss

$$\mathcal{L} = \frac{1}{2} \mathcal{L}_{\text{SupCon}} + \frac{1}{2} \mathcal{L}_{\text{Focal}}$$

Supervised contrastive [11]: same class attracts, different class repels; $\tau = 0.07$.

Focal loss [13]: down-weights easy examples; $\gamma = 2.0$, $\alpha = 0.75$.

→ 50 epochs, AdamW, effective batch 16; random flips, noise and intensity scaling applied to both scans.

Phase 2: population archive & retrieval

- Freeze the encoder; embed every **training** subject.
- Store $(\Delta v_i, y_i, C_i)$ in a **FAISS** exact inner-product index [10].
- Unit-norm vectors $\Rightarrow$ inner product = **cosine similarity**.

💡 Scales by swapping to an approximate index (e.g. IndexIVFPQ) — the workflow stays the same.

Top- k retrieval

$$\mathcal{N}_k(q) = \text{top-}k \left(\Delta v_q \cdot \Delta v_i \right)_{i \in \mathcal{D}}$$

$$k = 5$$

Δv_i : trajectory; y_i : label; C_i : age and sex of archived case i .

Phase 3: calibrated LLM reasoning agent

- **Llama-3-8B-Instruct**, 4-bit NF4 [8, 5], about 6 GB of GPU memory.
- Prompt: query age/sex + 5 retrieved cases.
- **Age-gap rule**: ignore matches more than $\delta = 15$ years apart.
- Output: ✓ SUCCESS or ✗ FAILURE + one sentence. Greedy decoding.

Calibrated probability

$$p_{\text{Oracle}} = 0.60 p_{\text{neighbor}} + 0.40 p_{\text{LLM}}$$

p_{neighbor} : share of the k neighbours with $y=1$

$p_{\text{LLM}} = 0.20$ for success, 0.80 for failure

Predict $\hat{y} = 1[p_{\text{Oracle}} > 0.5]$.

Experimental Setup

EPISURG [15]: public T1-weighted MRI of epilepsy surgery patients.

- **430** scans in total; **268** patients have both a pre- *and* a post-op scan.
- All **268** pass quality control.
- **215** with $y=0$ vs. **53** with $y=1$ (about **4:1**).

Proxy label — read results with care

Per-patient seizure outcomes (Engel scores) are **not public**.

- $y=0$: temporal resection (favourable proxy)
- $y=1$: extra-temporal (frontal, parietal, occipital) resection, hemispherectomy

 The encoder may learn **cavity location**, not true prognostic change.

Leak-free 5-fold stratified CV

- 5 folds, shuffled, seed 42; 4:1 ratio kept.
- 5 separate encoders: each never sees its test fold.
- FAISS index rebuilt per fold from non-augmented training subjects only.
- Every subject is tested exactly once.

Metrics (all higher is better)

- AUC-ROC — primary
- F1 (positive class)
- Sensitivity, specificity
- Balanced accuracy

Default threshold 0.50; also a grid search over [0.30, 0.50].

Seven methods, one cohort

Static encoders

Post-op scan only

- **M1** ResNet-50
- **M2** 3D-ViT

Siamese Δv classifiers

Same frozen trajectories

- **M3** Logistic regression
- **M3b** 2-layer MLP
- **M4** k -NN (cosine)
- **M6** Diversity ensemble

Proposed

Retrieval + LLM

- **M5** Neuro-Oracle

→ All **7** run on the same **268** subjects and the same folds. **M6** averages **4** models on Δv : ExtraTrees, two MLPs (different seeds), and a calibrated RBF-SVC.

Results

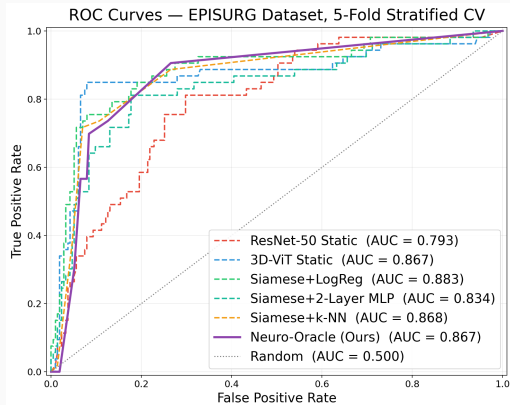
Main results

	Method	AUC	F1	Sens.	Spec.	Bal. Acc.
<i>Static</i>	ResNet-50 (M1)	0.793	0.462	0.396	0.921	0.659
	3D-ViT (M2)	0.867	0.775	0.811	0.930	0.871
<i>Siamese</i>	+ LogReg (M3)	0.883	0.629	0.849	0.791	0.820
	+ MLP (M3b)	0.834	0.628	0.811	0.809	0.810
	+ k-NN (M4)	0.868	0.717	0.717	0.930	0.824
	Ensemble (M6)	0.905	0.741	0.755	0.930	0.843
<i>Ours</i>	Neuro-Oracle (M5)	0.867	0.600	0.566	0.921	0.744

Table 1: **EPISURG**, $N=268$, 5-fold stratified CV, threshold **0.50**. All metrics: higher is better.

- ✓ Trajectory models lift sensitivity well above **M1** (**0.717–0.849** vs. **0.396**).
- M6** has the top AUC; the static **M2** has the top balanced accuracy.

ROC curves and Neuro-Oracle's decisions



ROC curves, six methods (M6 omitted for clarity).

Neuro-Oracle — Aggregated Confusion Matrix (5-Fold)

True label	Predicted label	
	Success (Engel I)	Failure (Engel II-IV)
Success (Engel I)	198	17
Failure (Engel II-IV)	23	30

M5 confusion matrix, aggregated over **5** folds.

- **198** of **215** class-0 cases correct (spec. **0.921**)
- **30** of **53** class-1 cases found (sens. **0.566**)

Moving the decision threshold

ID	Method	τ	AUC	Sens.	Spec.	Bal. Acc.
M3	Siamese + LogReg	0.50	0.883	0.849	0.791	0.820
M4	Siamese + k -NN	0.42	0.868	0.717	0.930	0.824
M5	Neuro-Oracle	0.34	0.867	0.698	0.916	0.807

Table 2: Threshold chosen by grid search to maximise balanced accuracy.

What changes for Neuro-Oracle

At $\tau = 0.34$: sensitivity $0.566 \rightarrow 0.698$, balanced accuracy $0.744 \rightarrow 0.807$, specificity barely moves ($0.921 \rightarrow 0.916$).

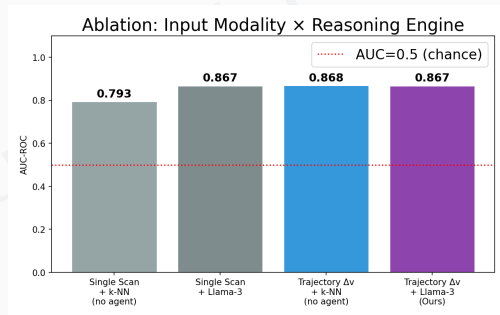


Threshold is tuned on the pooled held-out predictions.

Ablation: what each piece contributes

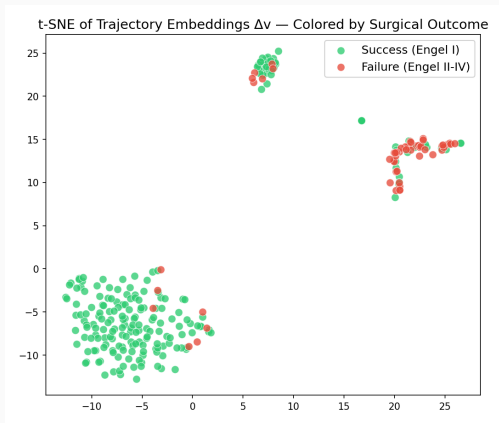
Configuration	AUC	Bal. Acc.
Random encoder	0.488	0.486
$k=1$ retrieval	0.741	0.741
No age filter	0.867	0.744
Full M5 ($k=5$, $\delta=15$)	0.867	0.744

- ✗ Untrained encoder: near chance.
- ✗ $k=1$: loses vote smoothing.
- Age filter: **no AUC change**; it shapes the reasoning.



Input (single scan vs. Δv) $\times$ engine (k -NN vs. **Llama-3**).

The trajectory space



t-SNE [17] of all 268 Δv vectors; green $y=0$, red $y=1$.

Retrieval quality

Top-5 retrieval fidelity $\uparrow$	83.0%
Mean cosine similarity (top-5) $\uparrow$	0.919
Calibration MAE $\downarrow$	0.189

- Class-0 cases form a large, clean cluster.
- Most class-1 cases fall in two smaller, **mixed** clusters.
- Fidelity: at least one correctly labelled case among the top 5.

Interpretability audit of the LLM agent

A real output (subject sub-0019)

"SUCCESS. The dominant surgical outcome of the remaining valid matches is SUCCESS, as 4 out of 5 patients with geometrically identical morphological atrophy trajectories and matching age and sex had successful outcomes, outweighing the single failure."

100.0%

zero-hallucination rate: no reply used facts
outside the prompt

73.9%

age-filter adherence: the age rule is followed in
most cases

 Audit-only research figures — not clinical safety guarantees. Zero hallucination reflects the tight prompt design.

Discussion & Conclusion

Interpretability vs. raw accuracy

M6 Diversity Ensemble

- ✓ Highest AUC: 0.905
- ✓ No LLM cost
- ✗ Bare score, no rationale

M5 Neuro-Oracle

- ✓ AUC 0.867, on par with M2 and M4
- ✓ Verdict + justification + the retrieved cases
- ✗ Conservative at $\tau = 0.50$ (sens. 0.566)

💡 A clinician can inspect the evidence behind each prediction before trusting it.

Limitations

- ✗ Proxy label: results reflect surgery type, not true seizure freedom.
- ✗ $N=268$: too small for tight confidence intervals.
- ✗ Single centre: no cross-site or cross-scanner evidence.

Next steps

- → Link scans to de-identified **Engel outcome** registries.
- → Test on **independent, multi-centre** cohorts.
- → Fine-tune the LLM on outcome-labelled **clinical notes**.

This work is an **architectural proof-of-concept**, ready for real outcome data.

Takeaways

1. ★ **Change over time carries signal** — trajectory models reach AUC **0.834–0.905** vs. **0.793** for the static ResNet-50 (a static 3D-ViT is competitive at **0.867**).
2. ★ **Neuro-Oracle** — matches trajectory classifiers at AUC **0.867** while explaining every prediction.
3. ★ **Grounded reasoning** — **100%** zero-hallucination rate under our audit protocol.

In one sentence

Neuro-Oracle turns MRI change into a searchable trajectory and explains its prognosis with similar past cases.

Thank you!

Questions?

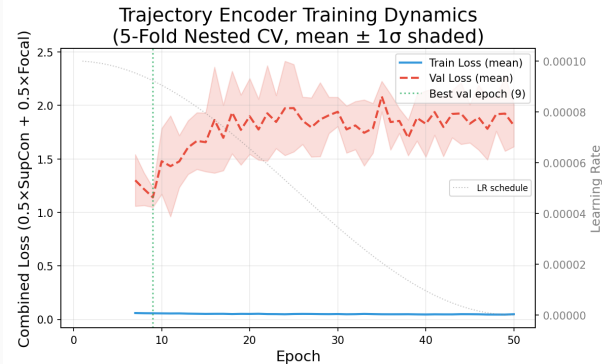
alexandera@gwu.edu

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Backup: encoder training dynamics



- Mean over 5 folds, $\pm 1\sigma$ shaded.
- Best validation loss at epoch 9.
- Validation loss rises after that, while training loss stays low.
- ⚠ Clear sign of **overfitting**: we keep the per-fold best checkpoint, not the last.

Backup: implementation details

Encoder

- **MONAI** 3D ResNet-50, layers [3, 4, 6, 3], from scratch
- AdamW, lr 10^{-4} , weight decay 10^{-4} , cosine schedule
- Batch 2×8 accumulation = 16; grad clip 1.0
- Trained on one **RTX 5060** GPU

Static baselines

- **M1/M2**: 25 epochs, class-weighted cross-entropy
- **M2** ViT: patch 16^3 , hidden 384, 6 layers, 6 heads

Agent

- Max 64 new tokens, greedy decoding